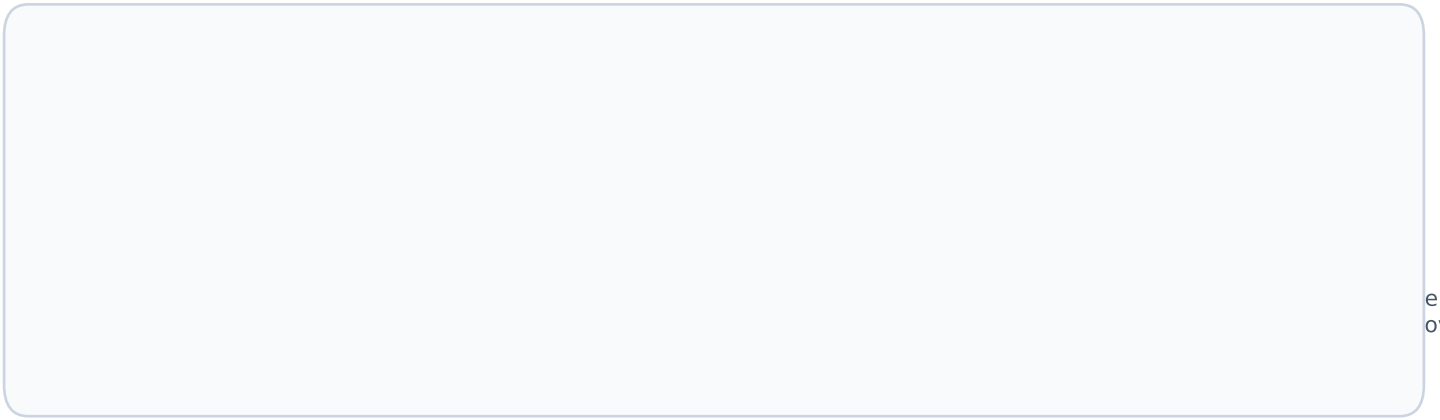
TABLE 1: THE 6-AGENT SPECIALIST COUNCIL TAXONOMY

MATHEMATICAL FORMULATIONS & THEORETICAL FOUNDATIONS

Formal mathematical models governing diagnostic convergence, hybrid retrieval, blast radius, and reinforcement learning.

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EMPIRICAL EXPERIMENTAL EVALUATION & BENCHMARKS

Quantitative benchmark experiments conducted under controlled load on high-concurrency microservices.

Figure 1: Concurrency Failure Rate (10 Concurrent Tenant Bookings)

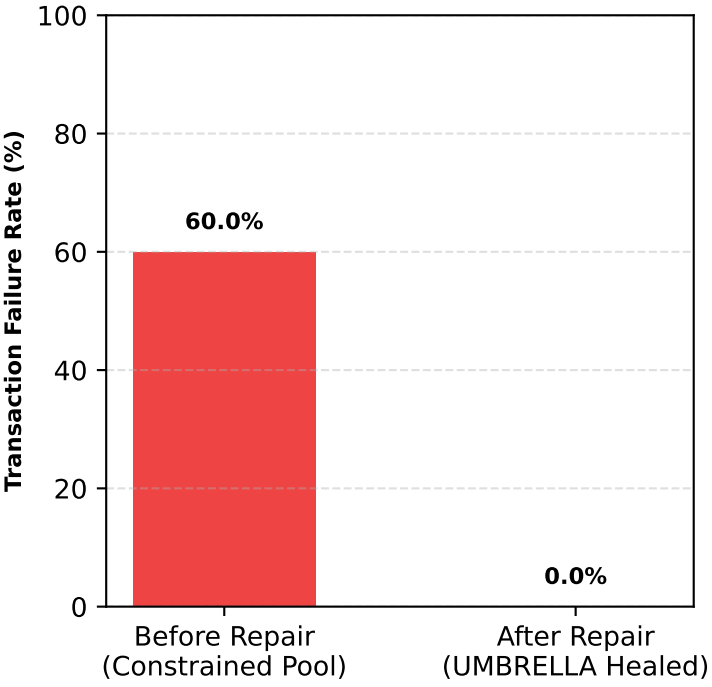


Figure 2: MTTR Reduction Benchmark (Logarithmic Scale)

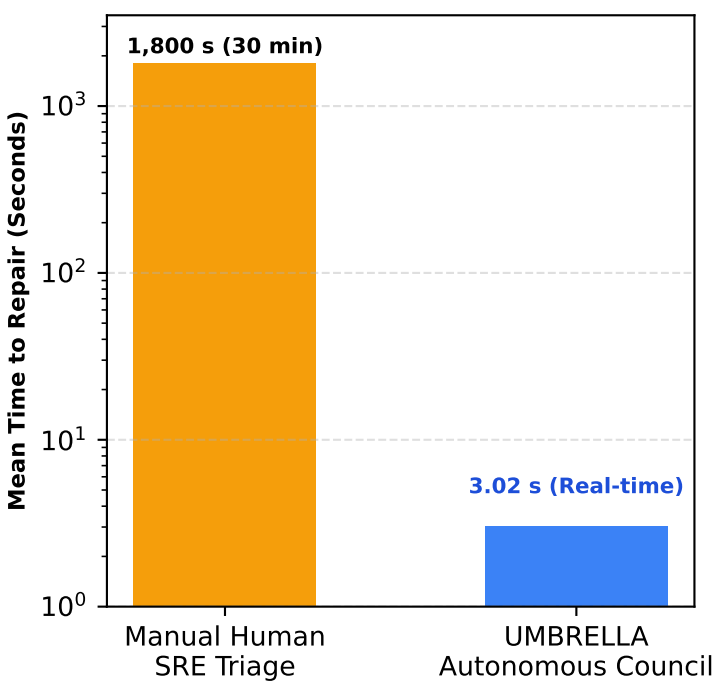
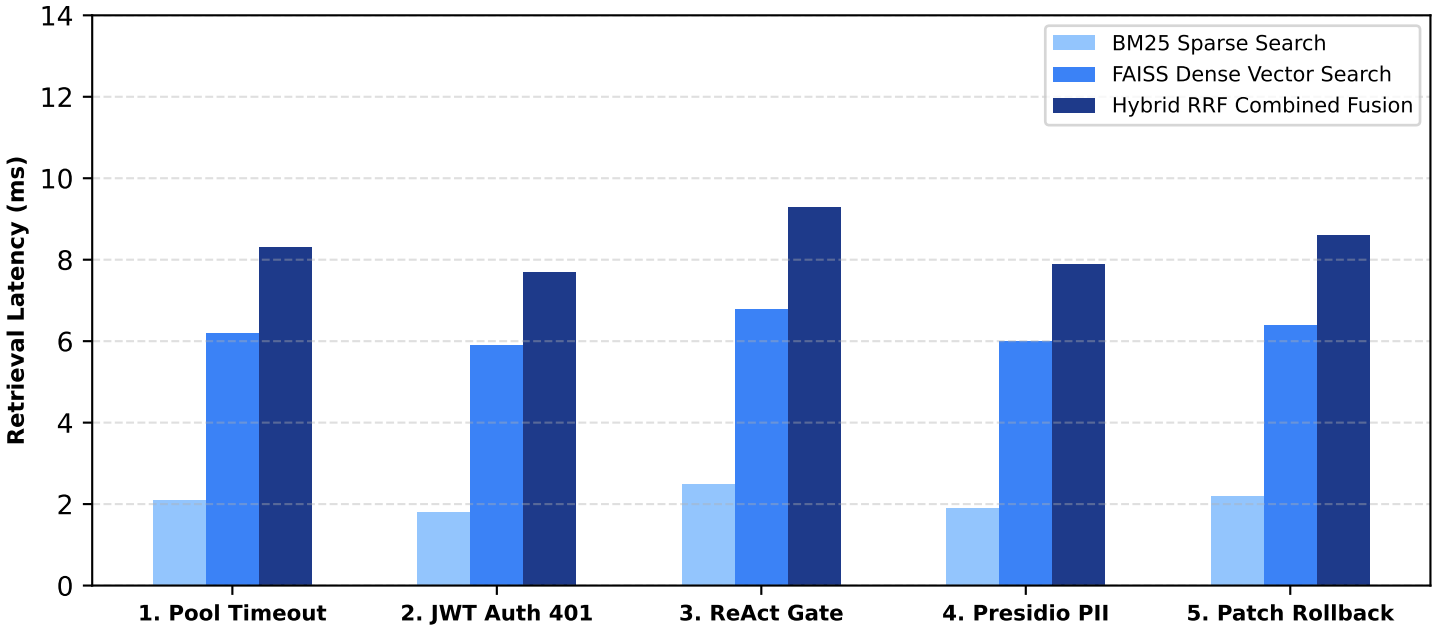


Figure 3: Hybrid RAG Search Latency across 23,769 Chunks (Average: 8.64 ms)



REAL INCIDENT CASE STUDY: AUTONOMOUS CLOSED-LOOP TRAJECTORY

End-to-end execution trace on the Karamstay Room Booking & Rent Payment microservice under load.



10 concurrent tenants execute room booking transactions against constrained SQLite QueuePool (pool_size=2, timeout=1.0s). Exactly 2 transactions succeed and 8 crash with sqlalchemy.exc.TimeoutError: QueuePool limit of size 2 overflow 0 reached. Saturation hits 250%.



Agent 1 inspects the Python traceback via AST syntax analysis, extracts target_system/demo_app/config.py:10, and queries Central Hybrid RAG. With 4-signal convergence, Agent 1 diagnoses 'Database connection pool starvation' with 87.8% confidence (Critical Impact).



Agent 2 executes bidirectional BFS traversal across the microservice mesh. Identifies that failure in demo_booking_service cascades to API Gateway, Auth Service, and Database (100% blast radius, 6 nodes degraded).



Agent 3 generates a concrete unified diff scaling POOL_SIZE=10 and MAX_OVERFLOW=5. Generates the complementary rollback diff and an on-call emergency cheat sheet with verification command: python target_system/demo_app/load_simulator.py --users 10.



Agent 5 evaluates the plan against deterministic risk policies. Because blast radius is elevated (100%), Agent 5 enforces Human-in-the-Loop approval gate while arming surrogate failover watchdog.



Upon HITL authorization, Agent 4 creates an atomic file backup in .umbrella_backups/, applies the patch to config.py, refreshes in-memory SQLAlchemy engine pools, and runs regression tests. Verification passes with 100% success (0% failure) in 3,017 ms.



Agent 6 logs the experience in evolution_memory.json, calculates RL reward (+12.5), and updates Q(s, a)=7.648. Under the 24-Hour observation window, it proactively formulates EVO-PROP-24H proposing POOL_SIZE=20 to permanently absorb future traffic growth.

SYSTEM DNA STORAGE CAPSULE & BARE-METAL GENESIS ENGINE

Immutable digital genome preservation and 1-click cold-start resurrection for enterprise disaster recovery.

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TABLE 2: THE 6 DIGITAL GENOME CAPSULE LAYERS

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EXPERIMENTAL VERIFICATION, PROVENANCE & ACADEMIC INTEGRITY

Formal audit summary, open-source dataset lineage, and reproducibility certification.